Primordial-Abundance Tests and Lithium Nucleogenesis

Detailed experimental programme, completed outcomes, inference limits and expanded research paths

Technical addendum to the cosmology research programme

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Current scientific position

The lithium discrepancy is real within the adopted direct-stellar anchor: standard-background LINX predicts Li-7/H near 5.6 x 10^-10 while the plateau likelihood is centred on 1.45 x 10^-10.

Deuterium and helium prevent the simple clock, baryon-density and selected-rate changes from lowering lithium enough.

The completed calculations robustly veto the tested FR-inspired clock proxy, the effective non-negative SU(2) expansion mapping and the literature-defined high-temperature SU(2)-CMB thermodynamic bridge as stand-alone lithium solutions.

They do not veto a future first-principles FR or SU(2) gauge-matter theory that predicts coherent changes to the thermal history and microphysics.

Abstract

Primordial lithium is mainly synthesised as beryllium-7 during standard Big Bang nucleosynthesis (BBN), then converted to lithium-7 by electron capture.

At the baryon density favoured by deuterium and the cosmic microwave background (CMB), modern networks predict an abundance roughly three to four times that inferred from warm, metal-poor stellar atmospheres.

This addendum documents five completed local experiments: an analytic emulator gate; a 30,503-point LINX sensitivity scan; a prospectively locked SU(2)-compatible completion gate and near-miss frontier; a 92-evaluation strict-tolerance, three-network validation matrix; and a separate 62-point SU(2)-CMB thermodynamic bridge.

The calculations are numerically stable, network-completeness corrections are far too small to solve lithium, and no simultaneous two-standard-deviation deuterium, helium and lithium pass exists.

The most valuable next step is a hierarchical BBN posterior that jointly marginalises nuclear-rate, observational and stellar-depletion uncertainties, cross-checked in an independent BBN code.

Model-specific FR and SU(2) work should proceed only after their thermal and microphysical maps are written explicitly enough to generate new rates and abundances without fitted ad hoc multipliers.

Headline findings

- **Completion and reproducibility.** The broad catalogue contains 30,503/30,503 successful unique states, with no unresolved point IDs; the targeted matrix completed 92/92 evaluations.
- **No abundance solution.** Registered and updated-current D+He+Li two-standard-deviation passes are both zero.
- **Numerical stability.** Strict-tolerance changes are at most 5.51 x 10^-4 in 10^5(D/H), 6.53 x 10^-5 in Y_p and 1.71 x 10^-3 in 10^10(Li/H).
- **Nuclear completeness.** The full network lowers lithium by only about 0.058 relative to the key network; the largest newly tested omitted-channel unit-pull response is 0.06079.
- **Primary statistical refinement.** The local deuterium theory scale, 0.02608 in 10^5(D/H), is comparable to the observational error and must be marginalised rather than omitted.

1. What Lithium Nucleogenesis Tests

BBN converts an initially free neutron-proton plasma into light nuclei as the temperature falls from the MeV scale to tens of keV. Weak interactions first set the neutron-to-proton ratio.

Nuclear assembly then waits until photodissociation can no longer destroy deuterium efficiently.

Once that deuterium bottleneck opens, most surviving neutrons are rapidly incorporated into helium-4; trace flows produce deuterium, helium-3, tritium, lithium-7 and beryllium-7.

Core abundance and freeze-out definitions

$$
\begin{aligned}
eta &= \text{n\_b / n\_gamma} \\
Gamma\_np(T\_f) &= \text{H(T\_f)} \\
n/p &= \text{exp[-Delta m\_np / T\_f] before neutron decay corrections} \\
Y\_p &= \text{2(n/p) / [1 + (n/p)] to leading order} \\
A(Li) &= \text{12 + log10[n(Li)/n(H)]}
\end{aligned}
$$

1.1 Dominant mass-seven pathways

Table 1. Reaction logic behind the cosmological lithium problem

Stage	Dominant reaction or process	Role in the final Li-7 abundance
Light-nucleus assembly	p(n,gamma)D; D(p,gamma)He-3; D(d,n)He-3; D(d,p)T	Sets the deuterium bottleneck and supplies He-3/T seeds.
High-eta production	He-3(alpha,gamma)Be-7	Dominant source of final mass seven at eta_10 near 6.
Low-eta production	T(alpha,gamma)Li-7	Direct Li-7 route; subdominant at the CMB baryon density.
Be-7 conversion	Be-7 + e- -> Li-7 + nu_e after BBN	Most primordial Li-7 is born as Be-7, not as Li-7.
Neutron destruction	Be-7(n,p)Li-7	Moves Be-7 into Li-7 while protons remain able to destroy Li-7.
Final destruction	Li-7(p,alpha)He-4	Removes Li-7 if conversion occurs early enough.

This topology explains why a useful lithium solution is difficult. Suppressing He-3(alpha,gamma)Be-7 or enhancing a Be-7 destruction channel must act strongly enough to reduce mass seven, yet must not disturb the deuterium-burning reactions that fix eta or the weak/expansion history that fixes helium.

A simple global clock change usually moves all three abundances together, and deuterium becomes the hard veto.

1.2 Likelihood quantities

Residual, rate-nuisance and stellar-depletion definitions

$$
\begin{aligned}
z\_X &= \text{[X\_pred - X\_obs] / sigma\_X} \\
chi2\_abund &= \text{z\_D\^{}2 + z\_He\^{}2 + z\_Li\^{}2} \\
q\_i &= \text{Normal(0,1) nuclear-rate pull} \\
R\_i(T,q\_i) &= \text{R\_i,0(T) exp[q\_i ln sigma\_i,exp(T)]} \\
f\_dep &= \text{(Li/H)\_BBN / (Li/H)\_surface}
\end{aligned}
$$

Interpretive distinction

A low observed stellar surface abundance can arise from low primordial production, depletion inside stars, errors in atmosphere/temperature modelling, or a mixture. A production calculation and a stellar observation model must therefore be fitted jointly before the lithium likelihood can be called primordial.

2. Experimental History, Programmes and Data

Table 2. Local primordial-abundance programme history

Date	Experiment	Programme/data	Decision delivered
16 Jul	Analytic FR scoping gate	analyze_bbn_lithium_fr_scoping_2026-07-16.py; fitted abundance emulator	Identified deuterium as the clock/eta veto and depletion as the viable nuisance direction.
16-17 Jul	FR/LINX key-network scan	analyze_bbn_lithium_linx_fr_network_2026-07-16.py; key_PRIMAT_2023	30,503 completed states; no joint abundance solution.
17 Jul	SU(2) completion gate	analyze_su2_bbn_lithium_gate_2026-07-17.py; immutable scan catalogue	Prospectively locked final 8,332 rows; zero expansion-only or rate-control passes.
17 Jul	Near-miss frontier	analyze_su2_bbn_lithium_frontier_2026-07-17.py	Quantified the residual lithium suppression/depletion burden after D+He gates.
17 Jul	Independent empirical audit	audit_bbn_lithium_linx_fr_scan_2026-07-17.py	Validated IDs and scores; exposed proposal-versus-prior and zero-threshold issues.
17 Jul	Validation matrix	validate_bbn_lithium_linx_matrix_2026-07-17.py; key/small/full_PRIMAT_2023	92/92 strict evaluations; numerical and network stability; D theory budget quantified.
17 Jul	SU(2)-CMB bridge	run_su2cmb_linx_thermal_bridge_2026-07-17.py; 62 eta states	Published asymptotic thermal benchmark strongly fails D+He and Li together.

2.1 Adopted observational anchors

Table 3. Registered and updated sensitivity anchors

Quantity	Registered anchor	Current sensitivity anchor	Function in the test
10^5(D/H)	2.508 +/- 0.029	2.533 +/- 0.024	Primary baryometer and strongest veto of clock/rate excursions.
Y_p	0.245 +/- 0.003	0.2458 +/- 0.0013	Constrains neutron-proton freeze-out, neutron lifetime and extra radiation.
10^10(Li-7/H)	1.45 +/- 0.25	Same direct-Gaussian control	Represents a metal-poor stellar surface plateau, not a depletion-corrected primordial likelihood.
eta_10	6.12 +/- 0.04	Matched CMB likelihood pending	Standard-cosmology baryon prior; only a compatibility stress for FR.
tau_n	879.4 +/- 0.6 s	878.4 +/- 0.5 s	Weak-rate calibration; update tested as sensitivity states.

The registered likelihood deliberately used transparent single-number anchors. That is adequate for a gate, but not for final inference.

In particular, a single Gaussian lithium datum suppresses the hierarchy among effective-temperature scales, one-dimensional versus three-dimensional atmospheres, local thermodynamic equilibrium versus non-local thermodynamic equilibrium (NLTE), evolutionary stage, metallicity, binary contamination and stellar depletion.

The current-anchor reweight is therefore a direction-of-sensitivity check, not a replacement likelihood.

2.2 Software and reproducibility

LINX (Light Isotope Nucleosynthesis with JAX) supplies differentiable background, weak-rate and nuclear-network integration. The broad run used the 12-reaction key_PRIMAT_2023 network at rtol=10^-5, atol=10^-9 and sampling_nTOp=150; interpretation-bearing validation rows used rtol=10^-7, atol=10^-11 and 300/600 interpolation samples. The audited environment used Python 3.11.15, NumPy 2.4.6, JAX/JAXLIB 0.4.28 and LINX revision ec2e9d2ca455e8204137e884da29f5dd13a638fa. Raw append-only attempts were retained, while final analysis reduced each point ID to its successful state.

3. Experiment A - Analytic FR Scoping Gate

3.1 Question and design

The first experiment asked whether a simple change to the early nuclear clock, baryon-to-photon ratio or lithium production could remove the discrepancy before committing to an expensive network calculation. It used analytic abundance fits as an emulator, not a reaction network. The scan covered 321 eta_10 values from 4 to 8, 241 clock factors from 0.85 to 1.15, 211 lithium-production suppression factors from 0.15 to 1.2 and 241 stellar-depletion factors from 1 to 5.

Scoping controls

$$\begin{aligned} S &= \mathrm{H\_BBN} / \mathrm{H\_SBBN} \\ Delta\ N\_eff &= (43/7)(\mathrm{S\hat{}2} - 1) \\ f\_prod &= \mathrm{(Li/H)\_produced} / \mathrm{(Li/H)\_standard} \\ f\_star &= \mathrm{(Li/H)\_produced} / \mathrm{(Li/H)\_surface} \end{aligned}$$

3.2 Outcome

Table 4. Scoping-gate model ranking

Model arm	Best chi2	Best eta_10	Best S	Interpretation
SBBN + CMB	19.9235	6.12	1.000	Lithium dominates the penalty.
Clock only	19.9235	6.12	1.000	The data return to the standard clock.
eta + clock + CMB prior	19.8273	6.1125	1.000	No meaningful clock displacement.
eta + clock free	17.9341	5.90	0.985	Small improvement, but CMB compatibility is surrendered.
Li production suppression	0.9575	6.12	1.000	Requires f_prod about 0.305.
Stellar depletion	0.9566	6.12	1.000	Requires f_star about 3.25.
Li as lower bound	0.9565	6.12	1.000	Removes the lithium penalty; not evidence for FR.

The emulator predicted standard Li-7/H = 4.72 x 10^-10 at eta_10=6.12, 3.26 times the adopted plateau centre. Deuterium inferred eta_10 about 6.04 while lithium treated as undepleted inferred eta_10 about 3.39.

The clock-only minimum was exactly S=1.

The experiment therefore did its job: it showed that a broad network calculation should focus on whether coherent microphysics can decouple mass-seven production from deuterium, while carrying a stellar-depletion branch as an explicit alternative.

> **Outcome boundary**
>
> This was a ranking and veto emulator. It neither solved a Boltzmann network nor tested a non-expanding FR thermal history.
>
> Its positive depletion arm is an astrophysical nuisance hypothesis, not a cosmological detection.

4. Experiment B - 30,503-Point FR/LINX Scan

4.1 Physical mapping actually solved

The network experiment replaced the emulator with LINX but retained the standard expanding, radiation-dominated thermal background.

FR was represented by an effective constant clock/radiation perturbation. Baryon density, neutron lifetime and selected nuclear-rate pulls were varied around standard values.

This is an FR-inspired sensitivity proxy, because it does not implement the research programme's non-expanding temperature-time relation or changing speed-of-light and mass scales.

Standard-background proxy map

$$
\begin{aligned}
S &= \text{H\_BBN / H\_SBBN} \\
\textit{Delta N\_eff\_init} &= \text{(43/7)(S\^{}2 - 1)} \\
\textit{Omega\_b h\^{}2} &= \text{0.02242 f\_eta} \\
\textit{eta\_10} &= \text{6.12 f\_eta} \\
\textit{tau\_n} &= \text{879.4 f\_tau seconds}
\end{aligned}
$$

Penalised profile objective

$$
\begin{aligned}
\textit{chi2\_scan} &= \text{chi2\_D + chi2\_He + chi2\_Li + chi2\_eta + chi2\_tau + Sum\_i q\_i\^{}2} \\
\textit{chi2\_X} &= \text{[(X\_pred - X\_obs)/sigma\_X]\^{}2}
\end{aligned}
$$

4.2 Catalogue construction and integrity

The designed catalogue combined fixed ladders with 30,000 random exploratory rows. It completed 30,503 successful unique states.

The append-only raw file retained 197 earlier failed attempts, so it contained 30,700 records; selecting the latest successful status per point reconstructed the exact target with no missing or unexpected IDs.

An independent audit found no semantic point-ID collisions, no random rounding collisions and score-reconstruction errors of only about 2.3 x 10^-12.

Table 5. Independent catalogue and proposal audit

Audit item	Observed value	Meaning
Successful unique states	30,503/30,503	Complete registered support.
Unresolved point IDs	0	No missing abundance result.
Exact duplicates after network normalisation	39	Five unavailable requested channels collapsed to fixed-rate rows.
Random proposal	q ~ N(0,1.35^2)	Exploration distribution was wider than the scored unit prior.
Threshold rule	q=0 when |q| <= 0.25	Creates an artificial point mass at zero.
Random all-zero rate rows	91/30,000	Expansion-only row counts are proposal-contaminated.

4.3 Outcome

Table 6. Broad-scan abundance frontier

Readout	D/H x 10^5	Y_p	Li/H x 10^10	Additional detail
Old best penalised direct-Li row	2.58506	0.248691	4.69411	chi2=215.256; pull penalty=22.845; He3aBe7g near -4 boundary.
D+He-constrained expansion-only floor	Passes gate	Passes gate	5.555	Residual depletion factor 3.83.
D+He-constrained modest-rate floor	Passes gate	Passes gate	5.113	Residual depletion factor 3.53.
D+He-constrained all-scan floor	Passes gate	Passes gate	4.741	Residual depletion factor 3.27.

No row entered the joint D+He+Li two-standard-deviation box. The best penalised row remained far above the lithium plateau despite an extreme He-3(alpha,gamma)Be-7 pull at the imposed boundary.

The scan therefore supplies a useful negative result: within its tested standard-background support, neither the clock proxy nor selected independent rate multipliers provide the required factor-three reduction while retaining deuterium and helium.

> **Inference boundary**
>
> The objective is a penalised profile score over a designed catalogue. It is not a posterior sample, a Bayesian evidence calculation or a goodness-of-fit chi-squared with one fixed number of degrees of freedom. Minima locate stress directions; they do not establish posterior support.

5. Experiment C - Preregistered SU(2) Gate and Frontier

5.1 Registration

When 22,171 of the 30,503 states were complete, the decision rule for the remaining 8,332 states was locked. This is a prospective completion gate for the unseen tail, not a blind preregistration of the whole scan.

The primary arm required non-negative extra radiation (S>=1), two-standard-deviation consistency with the standard CMB eta and neutron-lifetime priors, zero selected nuclear pulls, and separate two-standard-deviation passes in D, helium and lithium.

Modest and unrestricted rate-control arms were declared controls rather than SU(2) effects.

Locked completion-gate conditions

$$
\begin{aligned}
S &= \text{>= 1} \\
|f\_eta - 1| &= \text{<= 2(0.04/6.12)} \\
|f\_tau - 1| &= \text{<= 2(0.6/879.4)} \\
Sum\_i\ q\_i\hat{}2 &= \text{0 for the expansion-only arm} \\
|z\_D|, |z\_He|, |z\_Li| &= \text{each <= 2 for a candidate pass}
\end{aligned}
$$

5.2 Completed gate

Table 7. Locked SU(2)-compatible gate outcome

Scenario	Rows	D+He pass	D+He+Li pass	Minimum Li after D+He
SU(2) expansion only	25	16	0	5.555
SU(2) + modest rate controls	1,773	1,224	0	5.113
SU(2) + scanned rate controls	2,211	1,505	0	4.924
All scanned controls	30,503	4,715	0	4.741

5.3 Near-miss frontier

Frontier diagnostics

$$
\begin{aligned}
chi2\_D{+}He &= \text{z\_D^2 + z\_He^2} \\
f\_dep &= \text{(Li/H)\_BBN / mu\_Li} \\
f\_2sigma &= \text{(Li/H)\_BBN / (mu\_Li + 2 sigma\_Li)}
\end{aligned}
$$

Table 8. Remaining lithium burden after separate D and helium gates

Scenario	Li tension	Depletion to centre	Suppression to 2-sigma upper
Expansion only	16.42 sigma	3.831	2.849
Expansion + modest rates	14.65 sigma	3.526	2.622
Expansion + all selected rates	13.90 sigma	3.396	2.525
All scanned controls	13.16 sigma	3.270	2.431

The completion gate was decisively negative. The frontier adds scientific value by quantifying what a future microphysical model must accomplish: even the unrestricted explored controls require a further factor 2.43 suppression merely to reach the upper edge of the adopted two-standard-deviation lithium interval. Rate-control improvements cannot be labelled SU(2) unless a Lagrangian or finite-temperature calculation predicts their correlated signs and amplitudes.

6. Experiment D - 92-Evaluation Validation Matrix

6.1 Why this experiment was needed

The broad scan used economical tolerances and a key network.

The validation matrix tested whether its negative conclusion could be an artefact of solver accuracy, interpolation density, network truncation, an outdated neutron lifetime, fixed deuterium rates or omitted lithium/beryllium channels.

Ten interpretation-bearing states were rerun across key_PRIMAT_2023, small_PRIMAT_2023 and full_PRIMAT_2023.

Centred +/-1 sigma responses were computed for the three principal deuterium reactions and, in the full network, for five mass-seven channels absent from the key network.

Preregistered numerical settings

$$
\begin{aligned}
rtol &= 10\text{\^{}-7} \\
atol &= 10\text{\^{}-11} \\
N\_interp,primary &= 300 \\
N\_interp,check &= 600 \\
N\_evaluations &= 92/92 \text{ successful}
\end{aligned}
$$

6.2 Numerical and network outcomes

Table 9. Strict baseline by PRIMAT-2023 network

Network	Baseline D/H x 10^5	Baseline Y_p	Baseline Li/H x 10^10	Runtime
key	2.434293	0.2469465	5.695731	223 s
small	2.435004	0.2469494	5.638004	698 s
full	2.434914	0.2469325	5.637694	about 1,620 s

Table 10. Maximum absolute numerical and network differences

Validation contrast	Max |Delta D|	Max |Delta Y_p|	Max |Delta Li|
Strict key - stored loose key	0.0005507	0.00006532	0.001710
Sampling 600 - 300	0.0001157	0.00002017	0.000319
Small - key	0.0007384	0.000003743	0.05775
Full - key	0.0006485	0.00001483	0.05804

At the old best direct-lithium controls, the strict full network gives D/H=2.585689 x 10^-5, Y_p=0.248639 and Li-7/H=4.647982 x 10^-10, with registered direct-lithium chi2=210.57.

The corresponding strict key lithium prediction is 4.693676. Full-network completeness improves the number but is a precision correction, not a route to 1.45 x 10^-10.

6.3 Deuterium theory budget

Local independent-pull diagnostic in units of 10^5(D/H)

$$\begin{aligned} sigma\_D{,}theory &= \text{0.02608 at the full-network baseline} \\ sigma\_D{,}registered{,}combined &= \mathrm{sqrt}(0.029\text{^}2 + 0.02608\text{^}2) = 0.03900 \\ sigma\_D{,}current{,}combined &= \mathrm{sqrt}(0.024\text{^}2 + 0.02608\text{^}2) = 0.03544 \end{aligned}$$

The centred responses of D(p,gamma)He-3, D(d,n)He-3 and D(d,p)T imply a local deuterium theory scale almost as large as the registered observational error and larger than the current sensitivity error.

Quadrature is only a diagnostic because rate effects are non-linear and correlate D, helium and lithium. The publishable solution is to sample all rate pulls jointly within the abundance likelihood.

6.4 Omitted mass-seven channels

The five channels absent from the key network were tested in the full network.

The largest centred unit-pull lithium response came from Be-7(d,alpha)alpha+p: 0.06079 at baseline and 0.04640 at the best direct-lithium state. The next-largest omitted-channel response was only 0.00353.

These results agree with the broader laboratory conclusion that no conventional unmeasured reaction-rate change of plausible size supplies the needed factor-three reduction.

Validation decision

Do not repeat another broad blind selected-rate scan.

Use the full network for publication rows, marginalise the deuterium-rate nuisances, and direct new computation towards the stellar likelihood or a physically derived microphysical model.

7. Experiment E - SU(2)-CMB Thermodynamic Bridge

7.1 Identity and construction

This separately registered experiment implemented a literature-defined high-temperature SU(2)-CMB equation-of-state benchmark.

It must remain scientifically separate from the repository's late-time SU2/SU2R fluid.

Six extra gauge polarisations were treated as ideal ultra-relativistic modes.

Their energy density and heat capacity altered the LINX background and neutrino-temperature history; standard LINX weak rates and standard PRIMAT nuclear masses, binding energies, Q-values and reaction rates were retained.

All nuclear pulls were fixed to zero.

Published high-temperature benchmark relations

$$
\begin{aligned}
g\_gamma &= 2 \\
g\_YM &= 8 \\
xi &= \mathrm{g\_YM/g\_gamma} = 4 \\
nu\_CMB &= \mathrm{xi\hat{}(-1/3)} = 4\hat{}(-1/3) = 0.629960525 \\
T\_gamma(a) &= \mathrm{nu\_CMB\ T\_0\ /\ a} \\
rho\_YM &= \mathrm{xi\ rho\_gamma}
\end{aligned}
$$

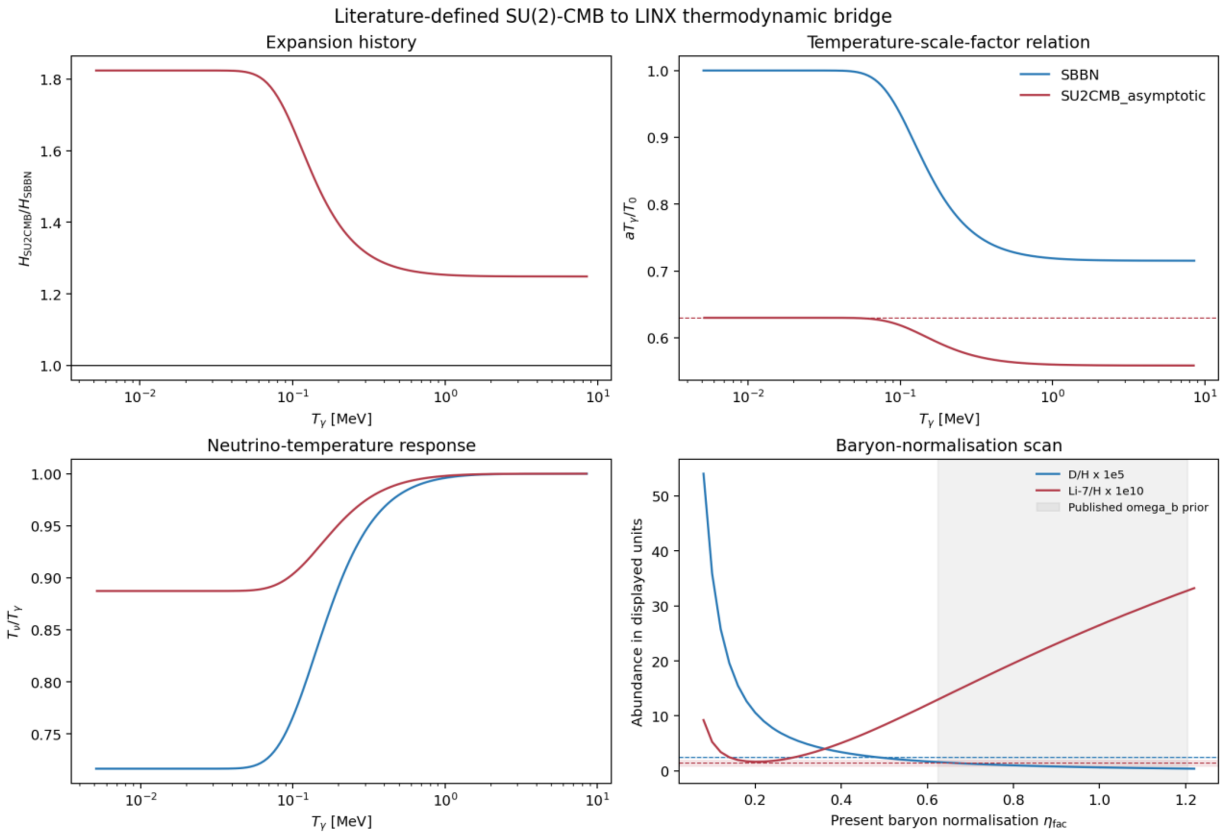

Figure 1. Literature-defined SU(2)-CMB thermodynamic bridge propagated through LINX.

At T_gamma=1 MeV the bridge gives H_SU2CMB/H_SBBN=1.25326.

The final LINX diagnostic N_eff is 20.373 rather than 3.0453.

These are not fitted lithium parameters; they follow from the implemented asymptotic mode count and temperature relation.

7.2 Abundance outcome

Table 11. Sixty-two-point SU(2)-CMB baryon-normalisation scan

Case	eta factor	D/H x 10^5	Y_p	Li/H x 10^10	chi2 D+He
SBBN reference	1.000	2.4337	0.24699	5.6973	6.99
SU(2)-CMB fixed reference eta	1.000	0.64441	0.31650	26.476	4,697.6
Entropy-cancel control	0.250	7.3229	0.30765	1.9191	28,003
Best D+He	0.480	2.5288	0.31214	7.6474	501.41
Best all abundances	0.460	2.7138	0.31187	6.9664	547.23
Best in published omega_b prior	0.62444	1.6109	0.31376	12.962	1,482.4

There are zero joint abundance passes. Reducing eta enough to lower lithium drives deuterium far too high; retaining the reference eta drives deuterium low, helium high and lithium extremely high.

The bridge therefore rejects the published asymptotic thermodynamic prescription as a stand-alone BBN solution under standard gauge-matter microphysics.

Claim boundary

The bridge is stronger than an arbitrary clock proxy because it propagates a published equation of state. It is still not a first-principles SU(2) gauge-matter BBN calculation: no SU(2)-specific electron radiative corrections, weak kernels, nuclear binding shifts, Q-value shifts or reaction-rate changes were supplied.

8. What Laboratory Experiments Already Tell Us

Because most final lithium is produced as Be-7, laboratory programmes have concentrated on its production and destruction.

The local full-network response matrix and modern direct measurements tell a consistent story: nuclear uncertainties remain important for precision BBN, especially deuterium, but the obvious mass-seven channels do not possess enough unmeasured leverage for a conventional factor-three lithium solution.

Table 12. Principal mass-seven laboratory/theory experiments

Experiment	Physical target	Outcome relevant to lithium	Research consequence
LUNA He-3(alpha,gamma)Be-7	Dominant Be-7 production at CMB eta	Direct low-energy measurements constrain the source reaction rather than revealing a factor-three overestimate.	Retain as a correlated rate nuisance; do not force a multi-sigma suppression without new data.
CERN n_TOF Be-7(n,p)Li-7	Main neutron-assisted Be-7 destruction	Measured from thermal energies to about 325 keV; the BBN window gives only a minor lithium improvement.	Use measured covariance/rate evaluation; no broad free multiplier.
CERN n_TOF Be-7(n,alpha)He-4	Direct Be-7 removal	First wide-range measurement found a minor BBN role and left the lithium problem unsolved.	Higher-energy completion is useful for precision, not presently a rescue priority.

Experiment	Physical target	Outcome relevant to lithium	Research consequence
Direct d+Be-7 measurements	Be-7 destruction through deuterons	Measured resonant strength reduces lithium modestly but does not solve the discrepancy.	Include in full-network covariance; avoid treating Be7daap as unconstrained.
2026 weak/radiative Be-7 calculation	In-situ e-capture, antineutrinos, positron decay and Be-7(p,gamma)B-8	Plasma screening/stimulated emission changes proton capture by about 1-3% near 87 keV; Auger-like contribution is tiny.	Implement as a precision update and compare to the existing unit-response bound.

8.1 Best nuclear next step

The prudent nuclear programme is a covariance update, not a blind rate hunt.

Assemble temperature-dependent posterior rate samples for the three deuterium reactions, He-3(alpha,gamma)Be-7, Be-7(n,p)Li-7, Be-7(n,alpha)He-4, Be-7(d,alpha)alpha+p and Li-7(p,alpha)He-4.

Propagate the same samples through LINX and one independent code.

This will answer whether the local linear theory budget remains valid away from baseline and will prevent a low-probability combination of rate tails from being misread as new physics.

Required covariance propagation

$$\begin{aligned} \textit{theta\_rate} &= \text{multivariate temperature-dependent rate parameters} \\ Y &= \text{F\_BBN(omega\_b, N\_eff, tau\_n, theta\_rate)} \\ \textit{p(Y|data)} &= \text{Integral p(data|Y,theta\_obs) p(theta\_rate) p(theta\_obs) dtheta} \end{aligned}$$

9. Stellar and Galactic Experiments

9.1 Stellar surface lithium

The direct lithium anchor is inferred from stellar absorption, usually the Li I 6707.8 angstrom feature.

Conversion to A(Li) depends on effective temperature, surface gravity, metallicity, line formation, three-dimensional atmospheric structure and NLTE corrections.

Atomic diffusion, gravitational settling, turbulent mixing, rotation, pre-main-sequence burning and later dredge-up can all separate present surface lithium from birth lithium.

The observed plateau's small scatter is informative, but does not prove zero depletion if the transport process is similarly regulated across the selected stars.

Table 13. Observation channels that separate primordial production from surface depletion

Observation	Experimental design	Reported outcome	Use in the next likelihood
Warm metal-poor plateau	High-resolution Li I spectroscopy of unevolved halo stars	Surface A(Li) near 2.1-2.3, below BBN+CMB near 2.7.	Fit raw measurements with survey and atmosphere offsets plus a depletion distribution.
2026 homogeneous NLTE sample	103 VMP/EMP stars down to [Fe/H]=-4.3	Slight positive plateau slope; possible extension below the claimed meltdown; evolutionary-state dependence.	Use star-level metallicity, temperature and evolutionary labels; do not compress to one Gaussian.
Lower red-giant branch	Post-dredge-up stars in field/cluster samples	A separate plateau near A(Li)=1.13 in the 2026 sample; abundance falls below 0.5 later.	Jointly calibrate dilution and transport against main-sequence stars.
Globular-cluster diffusion	Compare turn-off, subgiant and giant abundances in one population	Element-dependent trends support diffusion/mixing; inferred initial lithium is model-dependent.	Use multi-element trends to constrain the transport nuisance, not lithium alone.
SMC interstellar Li	Absorption in low-metallicity gas	Near the standard-BBN level, but chemical evolution and ionisation corrections remain.	Independent non-stellar-atmosphere control with explicit post-BBN enrichment.

9.2 Galactic lithium production

Lithium is also produced after BBN by novae through Be-7 transport and decay, by Galactic cosmic-ray spallation and alpha-alpha fusion, and by selected stellar sources.

These processes do not explain why old metal-poor surfaces are low by themselves, but they determine whether interstellar, meteoritic and young-star measurements can be used as primordial controls.

Nova observations and simulations currently show substantial yield scatter; chemical-evolution inference should therefore treat nova yields and delay-time distributions hierarchically rather than adopting one deterministic curve.

Conceptual chemical-evolution and surface-depletion decomposition

$$\begin{aligned} \mathit{Li\_ISM(t)} &= \text{Li\_BBN - astration} + \text{Li\_nova} + \text{Li\_GCR} + \text{Li\_stellar} + \text{inflow - outflow} \\ \mathit{Li\_surface} &= \text{Li\_birth / f\_dep(M, Z, age, rotation, mixing)} \end{aligned}$$

Observation-model priority

The present calculation has tested production much more thoroughly than observation. A star-level hierarchical lithium likelihood is now likely to improve research standing more than another large nuclear grid.

10. Expanded Research Paths

The following programme is ordered by evidential value, feasibility and ability to distinguish conventional systematics from genuinely new early-Universe physics.

10.1 Priority 0 - hierarchical CMB+BBN+stellar posterior

Use LINX's differentiability to sample a posterior over baryon density, effective radiation, neutron lifetime, every key-network rate pull, observational calibration terms and a physically constrained stellar-depletion model.

Replace the single lithium Gaussian with star-level or study-level data.

Run two lithium arms: a direct primordial plateau hypothesis and a depletion hierarchy.

Compare predictive performance rather than only minimum chi-squared.

Joint posterior target

$$
\begin{aligned}
\mathit{Theta\_BBN} &= \{\text{omega\_b, N\_eff, tau\_n, q\_1...q\_K}\} \\
\mathit{Theta\_star} &= \{\text{mu\_dep, sigma\_dep, slopes in T\_eff, [Fe/H], log g, study offsets}\} \\
\mathit{p(Theta|data)} &= \text{p(CMB|Theta\_BBN) p(D,He|Theta\_BBN) p(Li\_star|Theta\_BBN,Theta\_star) p(Theta)}
\end{aligned}
$$

Promotion gate: convergence with rank R-hat<=1.01, adequate effective sample sizes, prior-to-posterior overlap checks, posterior predictive calibration for each abundance, and a depletion posterior compatible with multi-element stellar transport rather than lithium alone.

10.2 Priority 0 - independent code reproduction

Reproduce a compact preregistered set of baseline, frontier and posterior-draw states in PRIMAT, PArthENoPE 3.0 or AlterBBN v2.

The goal is not exact equality because weak-rate and nuclear compilations differ; it is to show that the negative lithium conclusion and uncertainty budget survive an independent implementation.

Use full provenance for rate compilations and define tolerances before seeing the comparison.

Promotion gate: no code changes after unblinding; D and helium differences understood within documented physics choices; mass-seven conclusion unchanged across at least two solvers.

10.3 Priority 0 - stellar depletion and atmosphere hierarchy

Construct a curated star-level table containing equivalent width or fitted line abundance, T_eff method, log g, metallicity, evolutionary state, signal-to-noise, spectral resolution, NLTE/3D correction and membership/binarity flags.

Fit study-level temperature-zero-point offsets and depletion as functions of mass, metallicity and evolutionary state. Include globular-cluster turn-off-to-giant sequences to break the degeneracy between primordial abundance and surface transport.

Promotion gate: posterior predictive reproduction of the plateau width, metallicity trend, lower-red-giant plateau and multi-element diffusion signatures with no arbitrary star rejection after inspecting lithium.

10.4 Priority 1 - targeted nuclear covariance update

Replace independent scalar pulls with temperature-dependent rate posterior samples or a documented covariance basis. Recompute only the baseline, D+He frontier and posterior support region.

Add the 2026 Be-7 weak/radiative corrections as a separately switchable precision module.

This is a bounded experiment; the existing response matrix predicts that a rescue is unlikely.

Promotion gate: a proposed nuclear solution must pass D, helium and lithium simultaneously without requiring rate combinations outside laboratory posterior support.

11. Model-Specific and Alternative Paths

11.1 FR-native nucleosynthesis

A genuine FR calculation must replace the standard radiation-era clock rather than encode it through Delta N_eff.

The model must provide dimensionless, internally consistent functions for temperature, density, the speed of light, particle masses, weak interactions, binding energies and reaction detailed balance.

Only dimensionless predictions are observable; changing dimensional constants without specifying the measuring standards can be physically ambiguous.

Minimum FR-native input map

$$
\begin{aligned}
T(t) &= \mathrm{T\_FR(t)} \\
rho\_i(T) &= \mathrm{rho\_i,FR(T)} \\
c(T) &= \mathrm{c\_0\ F\_c(T)} \\
m\_j(T) &= \mathrm{m\_j,0\ F\_mj(T)} \\
eta(T) &= \mathrm{n\_b(T)/n\_gamma(T)} \\
Q\_np(T) &= \mathrm{m\_n(T)\text{-}m\_p(T)} \\
B\_A(T) &= \text{nuclear binding map} \\
Gamma\_np(T), R\_k(T) &= \text{FR weak and nuclear kernels satisfying detailed balance}
\end{aligned}
$$

Research sequence: derive the conservation laws and thermal clock; calculate weak freeze-out; propagate a coherent small basis of dimensionless coupling variations into masses and binding energies; validate the standard limit; then run D, helium and lithium together.

A matched FR CMB likelihood is ultimately required because the standard Planck baryon posterior may not apply.

11.2 SU(2) gauge-matter nucleosynthesis

The next SU(2) calculation should begin from an action-level finite-temperature model.

It must state whether the extra modes remain thermalised at BBN, how they couple to electrons and neutrinos, and whether electroweak/nuclear parameters shift.

The late-time SU2/SU2R fluid and SU(2)-CMB thermodynamics should remain separate until such a derivation proves a connection.

Promotion gate: recover the standard model continuously when the new coupling is removed; predict rather than fit the altered equation of state and microphysics; pass D+He before lithium is examined; and survive CMB N_eff and baryon-density constraints under the same model.

11.3 Energy injection and non-standard histories

Decaying or annihilating particles, photon cooling, primordial magnetic fields, spatially inhomogeneous baryon density and coherent variations of fundamental couplings can alter mass-seven production.

These paths are legitimate only as fully specified models.

Late neutron injection can destroy Be-7, for example, but generally raises deuterium; electromagnetic injection is constrained by photodissociation, CMB spectral distortion and entropy production.

The same D+He+Li likelihood and external constraints must be applied without choosing the favourable epoch after looking at the result.

Table 14. Alternative lithium paths and their unavoidable vetoes

Alternative	Potential lithium mechanism	Compulsory control
Late neutron injection	Convert Be-7 to Li-7 for proton destruction	Deuterium overproduction, helium and particle-decay limits.
Electromagnetic/hadronic decay	Photodissociate or transmute mass seven	D/He-3, CMB spectral distortion, entropy and gamma-ray constraints.
Photon cooling/entropy transfer	Change eta_BBN relative to eta_CMB	N_eff, neutrino temperature, CMB baryon density and D/H.
Varying dimensionless couplings	Shift Q_np, binding energies and Coulomb barriers coherently	Atomic-clock/quasar/CMB constraints and a microphysical sensitivity map.
Inhomogeneous BBN	Separate high- and low-density reaction flows	Diffusion scale, CMB isocurvature and volume-averaged D/He.

11.4 Galactic chemical-evolution closure

Fit BBN lithium, stellar astration, nova Be-7 yields, cosmic-ray production, inflow and outflow to a common time-metallicity model.

Use Li-6/Li-7 where available, open clusters, interstellar clouds, meteorites and nova gamma/optical Be-7 measurements.

This path does not change primordial production, but it tests whether one depletion/enrichment history can explain the plateau, the SMC gas abundance and the later Galactic rise without fine tuning.

12. Recommended Execution Plan

12.1 Immediate analyses

1. **Freeze the current abundance evidence.** Retain the 30,503-point catalogue, gate, frontier and 92-row matrix as immutable negative results; cite programme and LINX hashes.
2. **Build the star-level likelihood specification.** Define inclusion criteria, atmosphere corrections, study offsets, evolutionary classes and depletion parameterisation before importing lithium values.
3. **Assemble a nuclear covariance manifest.** Record source evaluation, temperature grid, covariance representation and code mapping for each rate; distinguish measured posterior support from exploratory multipliers.
4. **Register the cross-code anchor set.** Use the strict baseline, best direct-Li controls, D+He frontier states, neutron-lifetime update and a small number of posterior draws.

12.2 Next substantial computation

Run the hierarchical LINX posterior first.

It directly addresses the main limitation exposed by the validation matrix, uses LINX's differentiability and can quantify whether the data favour stellar depletion, rate tails or a residual cosmological discrepancy.

Begin with standard BBN plus the registered/current anchor ladder, then add the star-level hierarchy.

Reserve FR-native and SU(2) runs until their physical maps exist; more computation with the present clock proxies will mostly repeat the same veto.

Table 15. Staged research programme and decision rules

Stage	Primary output	Stop/go rule
H1: Standard BBN posterior	Joint omega_b, tau_n and all key-rate posterior; D/He predictive checks	Proceed only after convergence and independent likelihood reconstruction.
H2: Stellar hierarchy	Primordial Li posterior and depletion distribution	Promote depletion only if stellar evolutionary controls are reproduced.
H3: Cross-code anchors	LINX versus PRIMAT/PArthENoPE/AlterBBN differences	Resolve discrepancies larger than the declared theory budget.
H4: FR-native kernel	T(t), weak freeze-out and coherent nuclear sensitivity basis	No abundance scan until detailed balance and standard-limit tests pass.
H5: SU(2) gauge-matter kernel	Finite-temperature EoS and coupling map	No lithium claim unless D+He and matched CMB constraints pass first.

12.3 Predeclared claim ladder

- **Level 0 - numerical result.** A programme completed and the stored arithmetic is reproducible.
- **Level 1 - sensitivity result.** A parameter direction changes lithium but may use proxy physics or profile minima.
- **Level 2 - model candidate.** A specified model passes D, helium and lithium with all nuisance priors and external constraints.
- **Level 3 - robust inference.** The result survives posterior predictive checks, prior sensitivity and an independent BBN implementation.
- **Level 4 - research claim.** Independent data or an independent group reproduces the effect under a preregistered analysis.

Recommended present wording

The completed standard-background LINX programme finds no joint primordial-abundance solution from the tested clock, baryon-density, neutron-lifetime and nuclear-rate controls.

Numerical convergence and full-network checks strengthen this negative result.

The leading unresolved question is whether a hierarchical stellar-depletion likelihood reconciles the lithium observations with standard BBN; FR and SU(2) require separate first-principles microphysical construction before further abundance claims.

13. Sources and Evidence

13.1 Local evidence archive

FR emulator gate: github_export/results/2026-07-16/bbn_lithium/bbn_lithium_fr_scoping_report.md

Detailed audit: github_export/results/2026-07-17/bbn_lithium/fr_linx_analysis_program_detailed_audit_2026-07-17.md

Empirical scan audit: github_export/results/2026-07-17/bbn_lithium/fr_linx_scan_empirical_audit_2026-07-17.md

Anchor sensitivity: github_export/results/2026-07-17/bbn_lithium/fr_linx_anchor_sensitivity_report_2026-07-17.md

SU(2) registration and gate: github_export/results/2026-07-17/bbn_lithium/su2_bbn_lithium_completion_gate_preregistration.md and su2_bbn_lithium_gate_report_2026-07-17.md

Near-miss frontier: github_export/results/2026-07-17/bbn_lithium/su2_bbn_lithium_frontier_report_2026-07-17.md

Validation interpretation: github_export/results/2026-07-17/bbn_lithium/fr_linx_validation_matrix_interpretation_2026-07-17.md

SU(2)-CMB bridge: github_export/results/2026-07-17/bbn_lithium/su2cmb_linx_thermal_bridge_report_2026-07-17.md

Programmes: github_export/code/bbn_lithium/

13.2 Primary scientific sources

Giovanetti et al., LINX: A Fast, Differentiable, and Extensible Big Bang Nucleosynthesis Package, arXiv:2408.14538. https://arxiv.org/abs/2408.14538

Pitrou et al., Precision Big Bang Nucleosynthesis with Improved Helium-4 Predictions (PRIMAT), arXiv:1801.08023. https://arxiv.org/abs/1801.08023

Gariazzo et al., PArthENoPE Revolutions, arXiv:2103.05027. https://arxiv.org/abs/2103.05027

Arbey et al., AlterBBN v2, arXiv:1806.11095. https://arxiv.org/abs/1806.11095

Kislitsyn et al., New Precise Primordial Deuterium Determination, arXiv:2401.12797. https://arxiv.org/abs/2401.12797

Aver et al., LBT Y_p Project IV, arXiv:2601.22238. https://arxiv.org/abs/2601.22238

Yeh et al., LBT Y_p Project V, arXiv:2601.22239. https://arxiv.org/abs/2601.22239

Damone et al., Be-7(n,p)Li-7 at n_TOF, arXiv:1803.05701. https://arxiv.org/abs/1803.05701

Barbagallo et al., Be-7(n,alpha)He-4 at n_TOF, arXiv:1606.09420. https://arxiv.org/abs/1606.09420

Taioli et al., Be-7 Weak and Radiative Transition Rates in BBN, arXiv:2601.12438. https://arxiv.org/abs/2601.12438

Yan et al., Homogeneous NLTE Lithium in 103 Very Metal-Poor Stars, arXiv:2605.19334. https://arxiv.org/abs/2605.19334

Aguilera-Gomez et al., Lithium in the Lower Red-Giant Branch of Five Globular Clusters, arXiv:2109.13951. https://arxiv.org/abs/2109.13951

Howk et al., Interstellar Lithium in the Small Magellanic Cloud, arXiv:1207.3081. https://arxiv.org/abs/1207.3081

Clara and Martins, Primordial Nucleosynthesis with Varying Fundamental Constants, arXiv:2001.01787. https://arxiv.org/abs/2001.01787

Yamazaki et al., Photon Cooling, X-Particle Decay and a Primordial Magnetic Field, arXiv:1407.0021. https://arxiv.org/abs/1407.0021

SU(2)-CMB high-temperature relation and cosmological implementation: arXiv:1712.08561 and arXiv:1810.01253. https://arxiv.org/abs/1712.08561 ; https://arxiv.org/abs/1810.01253

13.3 Acronyms

Table 16. Acronym key

Acronym	Meaning	Acronym	Meaning
BBN	Big Bang nucleosynthesis	CMB	Cosmic microwave background
D/H	Deuterium-to-hydrogen number ratio	Y_p	Primordial He-4 mass fraction
FR	Research programme's non-standard cosmological framework	LINX	Light Isotope Nucleosynthesis with JAX
NLTE	Non-local thermodynamic equilibrium	SBBN	Standard Big Bang nucleosynthesis
VMP/EMP	Very/extremely metal-poor	GCR	Galactic cosmic ray
PRIMAT	Precision BBN code/rate framework	SU(2)-CMB	Literature-defined SU(2) CMB thermodynamics

Conclusion

The primordial-abundance programme has progressed from a qualitative lithium question to a reproducible set of numerical vetoes and uncertainty diagnostics.

The result is not a solution, but it is a stronger research position: simple FR clock changes, effective SU(2) extra radiation and the asymptotic SU(2)-CMB bridge do not preserve D and helium while fixing lithium; solver and network errors are too small to explain that failure; and the main missing conventional ingredients are joint deuterium-rate marginalisation and a serious stellar observation/depletion likelihood.

Those two tasks, followed by independent code reproduction, should precede further broad parameter searches.

Model-specific FR or SU(2) nucleogenesis remains worth investigating only through explicit thermal and microphysical derivations that make new, jointly testable abundance predictions.